

NEIL JOHARI

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EXPERIENCE

Roblox [San Mateo, CA]

July 2021 – present

Staff Software Engineer

Performance & Compression

- Co-led the largest 2024 time-to-interactivity win: cut feature-flag payloads **90%** across the game engine's four native clients (macOS, Windows, iOS, Android), CDN, and backend for up to **1s faster p95** and a measured lift in topline revenue metrics (zstd shared dictionaries)
- Conceived and led a managed compression platform: won SVP-level cross-org funding and validated **40-60% data-size reduction** across 10 production use cases (shared dictionaries, stateless API)
- Delivered the first production use case, compressing translation data: **50-60% memory reduction**, halving the Redis cluster; building the general library with multi-cloud redundancy and no single point of failure (Rust; C#/Python interop)
- Contributed fixes upstream to [zstd](#) and [OpenZL](#) (with their creator Yann Collet) and to [curl](#) (C, C++)

Content Safety & Moderation

- Led ingestion infrastructure and 3 engineers for a high-priority, six-month company-wide platform-safety effort with weekly SVP reviews: **250k docs/sec** across a **>10B-document corpus** (OpenSearch, CDC)
- Engineered a realtime streaming re-moderation pipeline, reprocessing content on access in response to emerging real-world events while balancing CDN and ML cost; dedup and stateful filtering cut **150k qps to 20 qps** (Flink, VLMs)
- Architected a 3-team system reindexing content on moderation events so takedowns propagate to search and discovery

Content Understanding & ML

- Designed and trained a novel SE(3)-equivariant transformer for perceptual hashing over 3D point clouds, beating prior published methods on internal benchmarks (PyTorch, Ray)
- Built agentic content-interrogation workflows over 3D assets: extracting structure and quality signals, analyzing embedded scripts, detecting malicious assets, and rendering screenshots for VLMs and embeddings
- Created a declarative framework triggering Temporal workflows from Kafka events, powering content copy-detection (Kafka Connect SMTs, DuckDB)

Search & Relevance

- Tech Lead for a two-quarter, 6-team content search re-architecture: **50-60% lower latency** (250-500ms p99), 18 services collapsed to 3, experiment setup cut from weeks to days
- Shipped multimodal semantic search (CLIP, SigLIP, CLAP, Milvus) for **+8.5% deep engagement**, with custom blending and re-ranking; set the team's relevance-evaluation standard (NDCG)
- Created facet generation for search covering **~70% of head queries**, with semantic fuzzy matching and precomputed facets (LLMs)

Core Platform & Reliability

- Proposed and shipped backend feature flagging (**40+ services**) and automatic distributed tracing (**~2.2k microservices**); presented both as company-wide tech talks
- Enabled HTTP/2 in the game engine: integrated nghttp2 and contributed custom memory allocators upstream to curl; partnered with the CDN team on the rollout, cutting p50 client TCP connections **60 to 20** (C, libcurl)
- Served as the platform's primary Incident Manager On-Call (IMOC) for a two-quarter rotation, coordinating incident response escalated to VP and C-suite levels

University of Michigan [Ann Arbor, MI], Instructional Aide & Research Assistant

Sept 2018 – May 2021

- Led a lab section for EECS 280 (~4.6k students/semester); patched the [autograder](#), automated weekly outreach to ~1.2k students, and published [Claw](#), an office-hours debugging CLI (Vue, Python, Ruby)

EDUCATION

University of Michigan [Ann Arbor, MI]

Sept 2018 – May 2021

Bachelor of Science in Engineering in Computer Science, summa cum laude

PROJECTS

- [urhired.ai](#): Built and self-hosted a realtime voice-to-voice LLM agent with live transcription and an LLM-visible whiteboard, deployed as an AI system-design mock interviewer (React, Python, Terraform, Ansible)
- BRADMEN Search Engine: Architected a distributed web search engine: crawler, indexer, query compiler, ranker, and frontend, crawling **~125M pages** and indexing **~25M** (C++)
- [AutoBroadcaster+](#): Authored a Minecraft scheduled-broadcasting plugin with **229k+ downloads** (Java, Kotlin)

ADDITIONAL

- Languages: C#, Python, C/C++, Java, Rust, TypeScript/JavaScript, SQL, GraphQL
- Systems & Data: Terraform, Ansible, Kafka, Temporal, OpenSearch, Airflow, Tecton, PyTorch, Ray, DuckDB, GitHub Actions, Protobuf, React, Rails, PostgreSQL
- Focus Areas: Search & Relevance, Applied ML, Data Compression, Reliability, Distributed Systems, Platform/Infra